Noise Analysis — 2um

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PAGE 1 — FILE STATISTICS

Folder:           data/dataset_hamming_c1/test/2um
Files:            240
Signal length:    16384–16384 samples
Dtype:            float64

Aggregate noise statistics:
  mean(σ):        0.116761
  std(σ):         0.050236
  CV(σ):          0.4302
  mean(RMS):      0.136105
  mean(kurtosis): -0.3088  (0 = Gaussian)

  mean(DC):       0.002624
  std(DC):        0.071754
  Amplitude range: [-0.8313, 1.2452]
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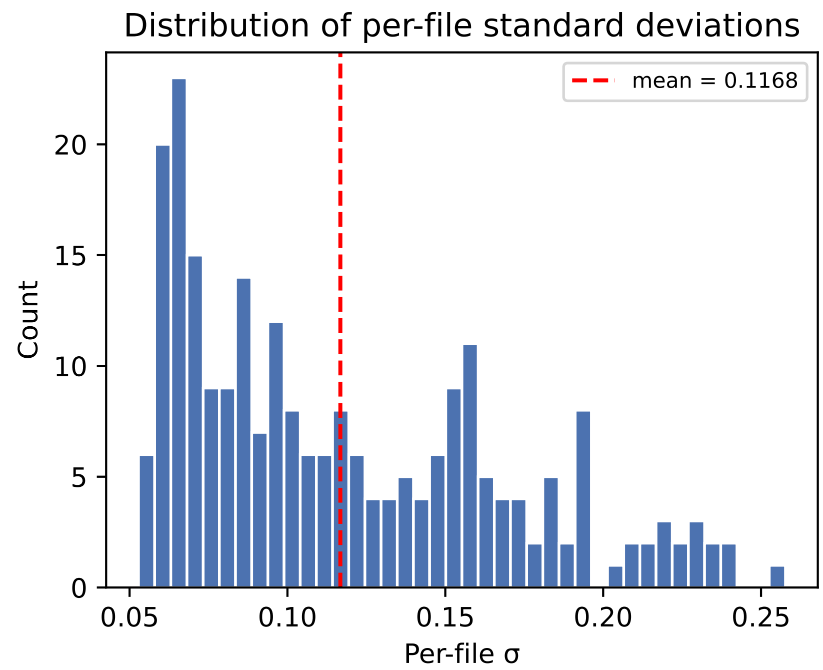
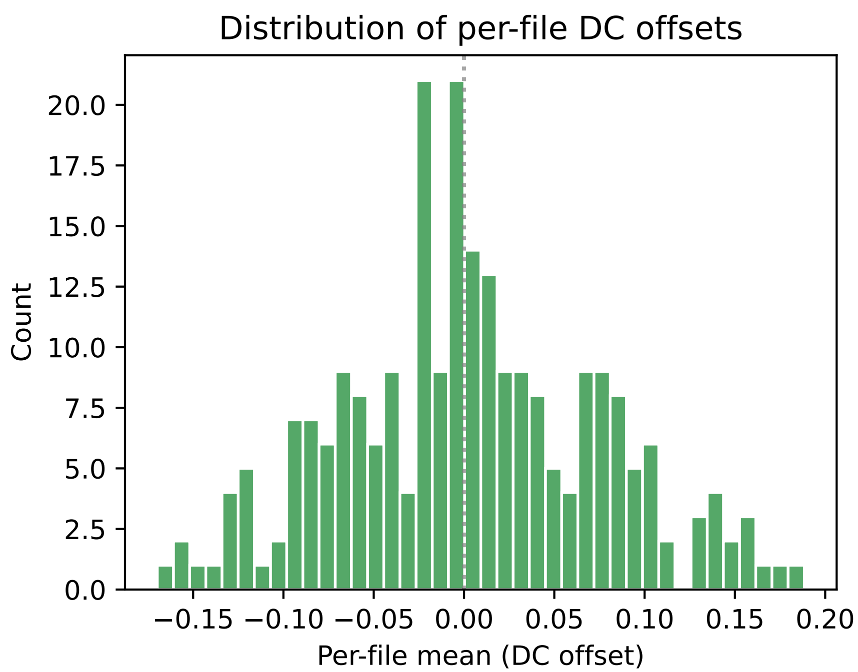
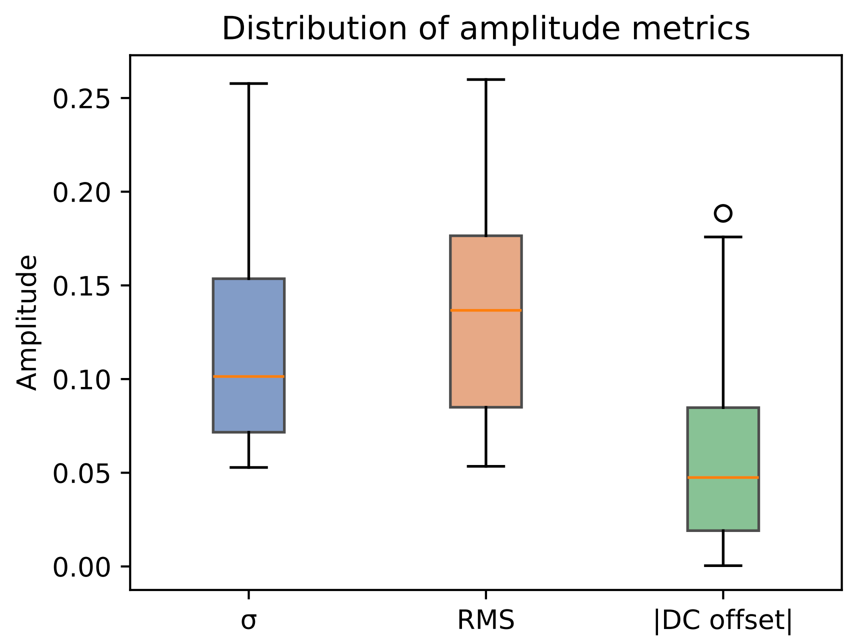
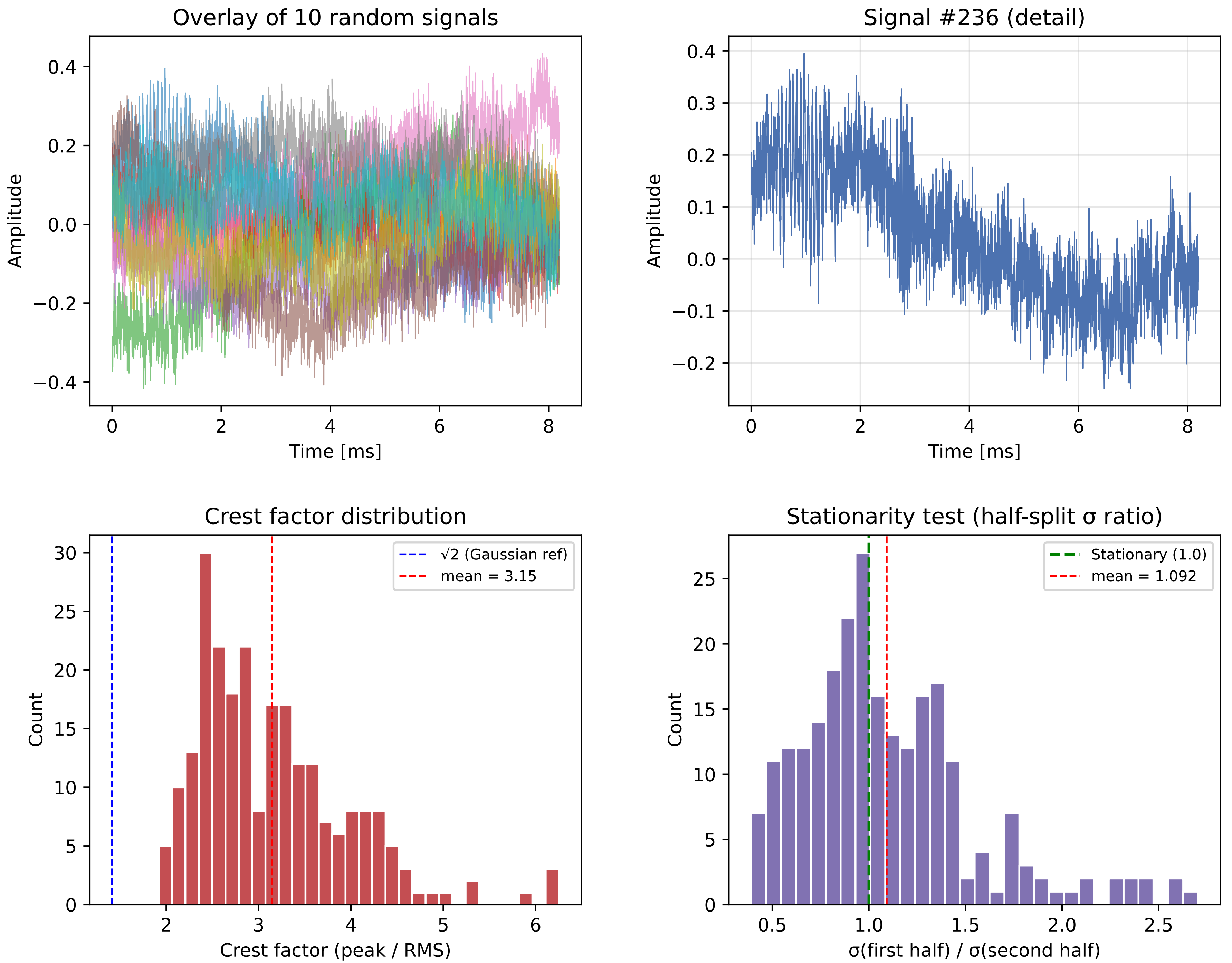


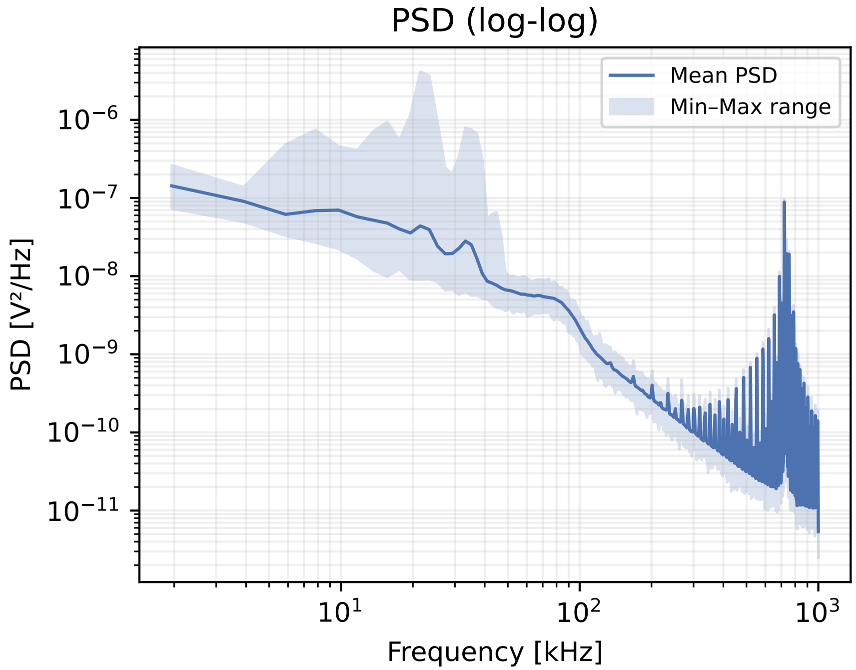


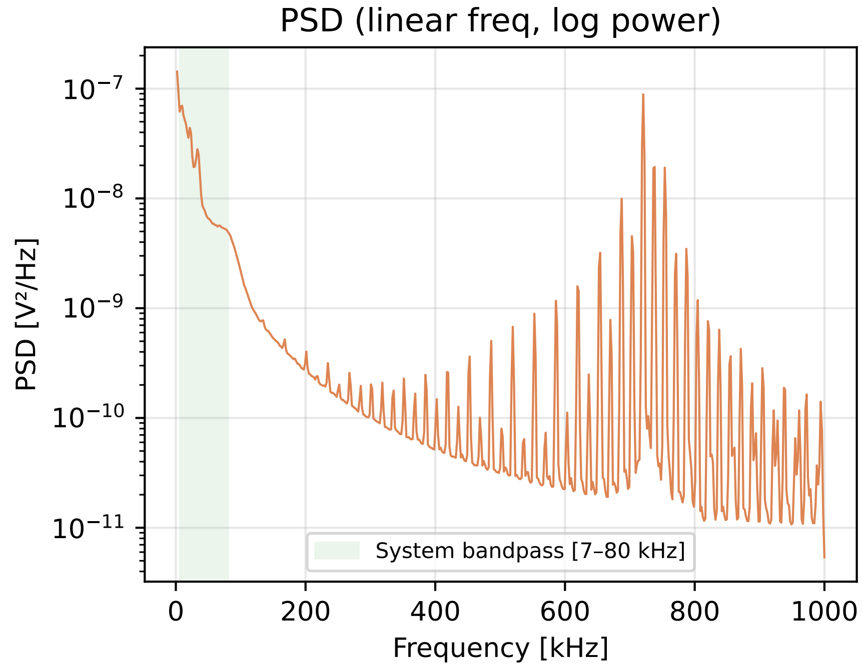


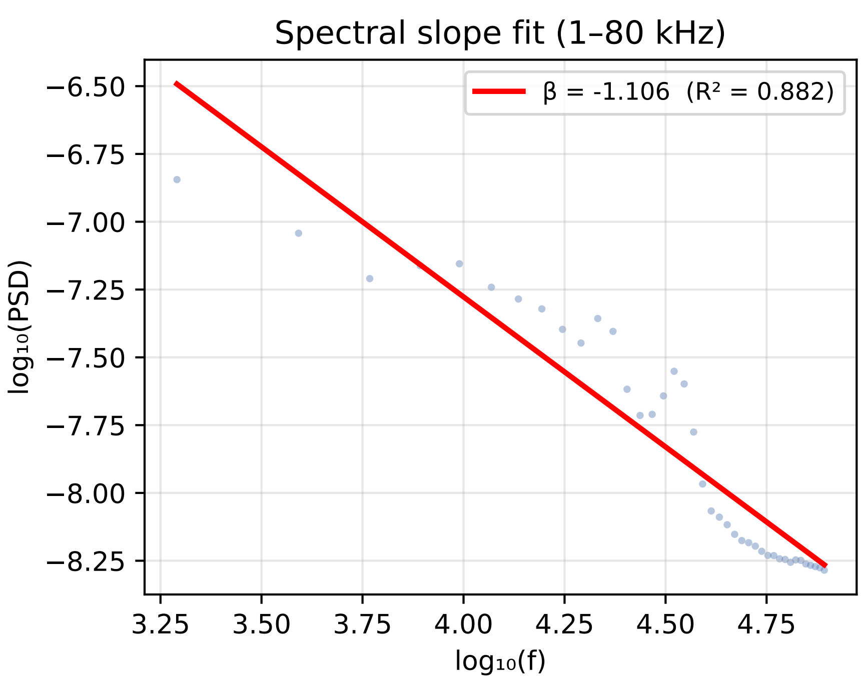
Time-Domain Analysis — 2um



Power Spectral Density — 2um





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SPECTRAL ANALYSIS SUMMARY

Spectral slope β:      -1.1064
Fit quality R²:        0.8824
Spectral flatness:     0.0552  (1.0 = white)

Slope interpretation:
  β ≈  0 → White noise    (flat PSD)
  β ≈ −1 → Pink / 1/f     (equal energy per octave)
  β ≈ −2 → Brown / 1/f²   (random walk)

Classification:        Pink (1/f)
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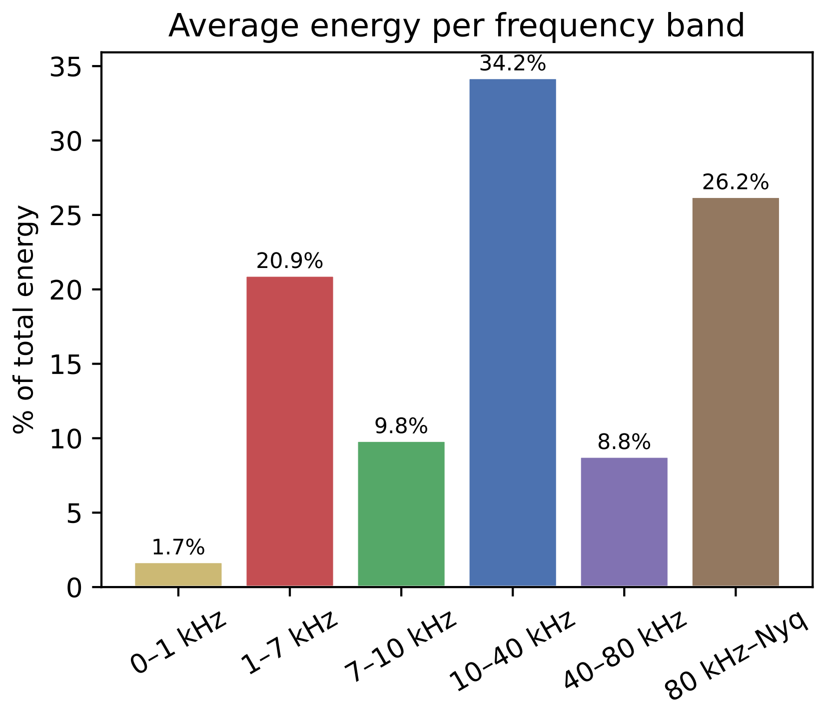
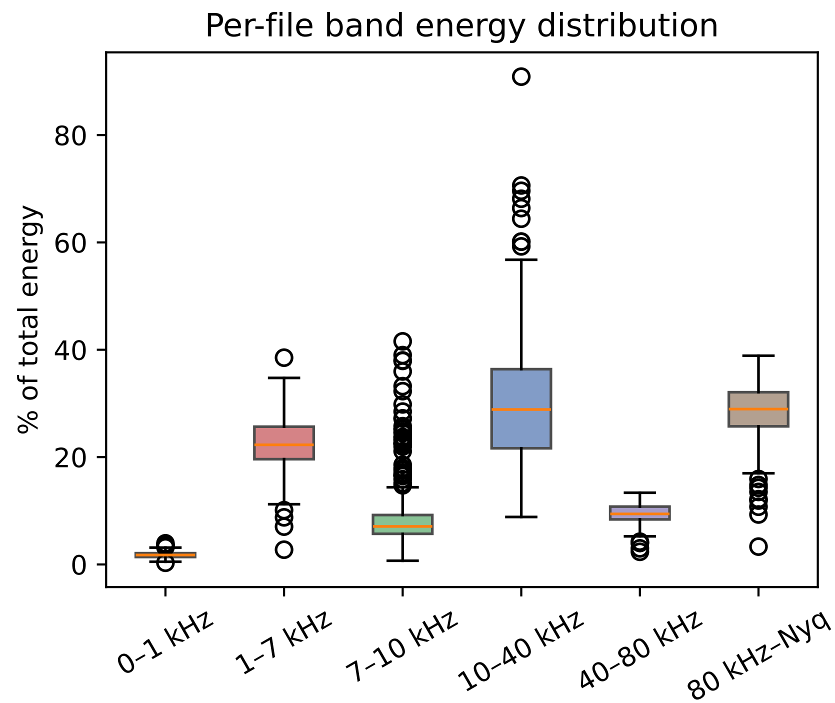
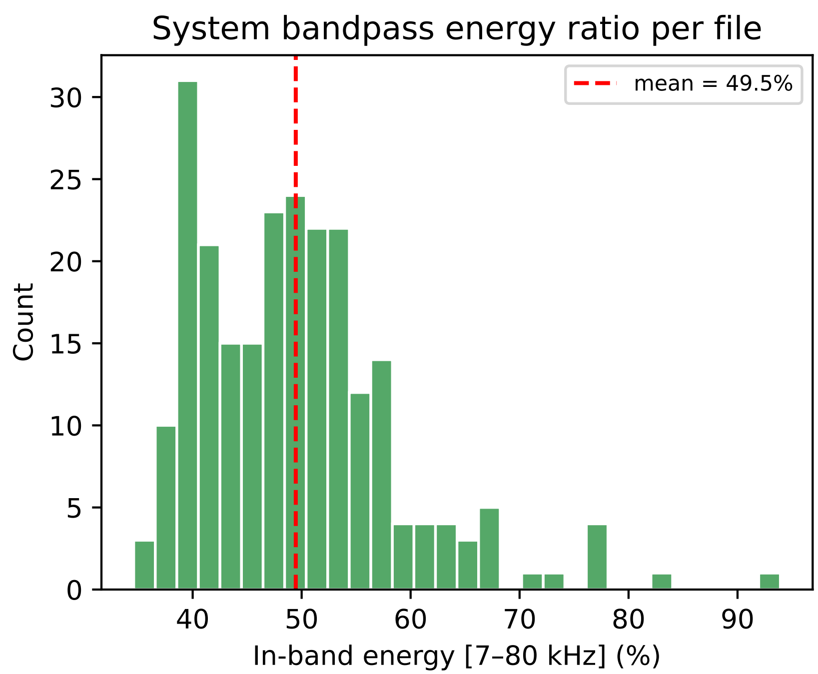
Frequency Band Energy — 2um



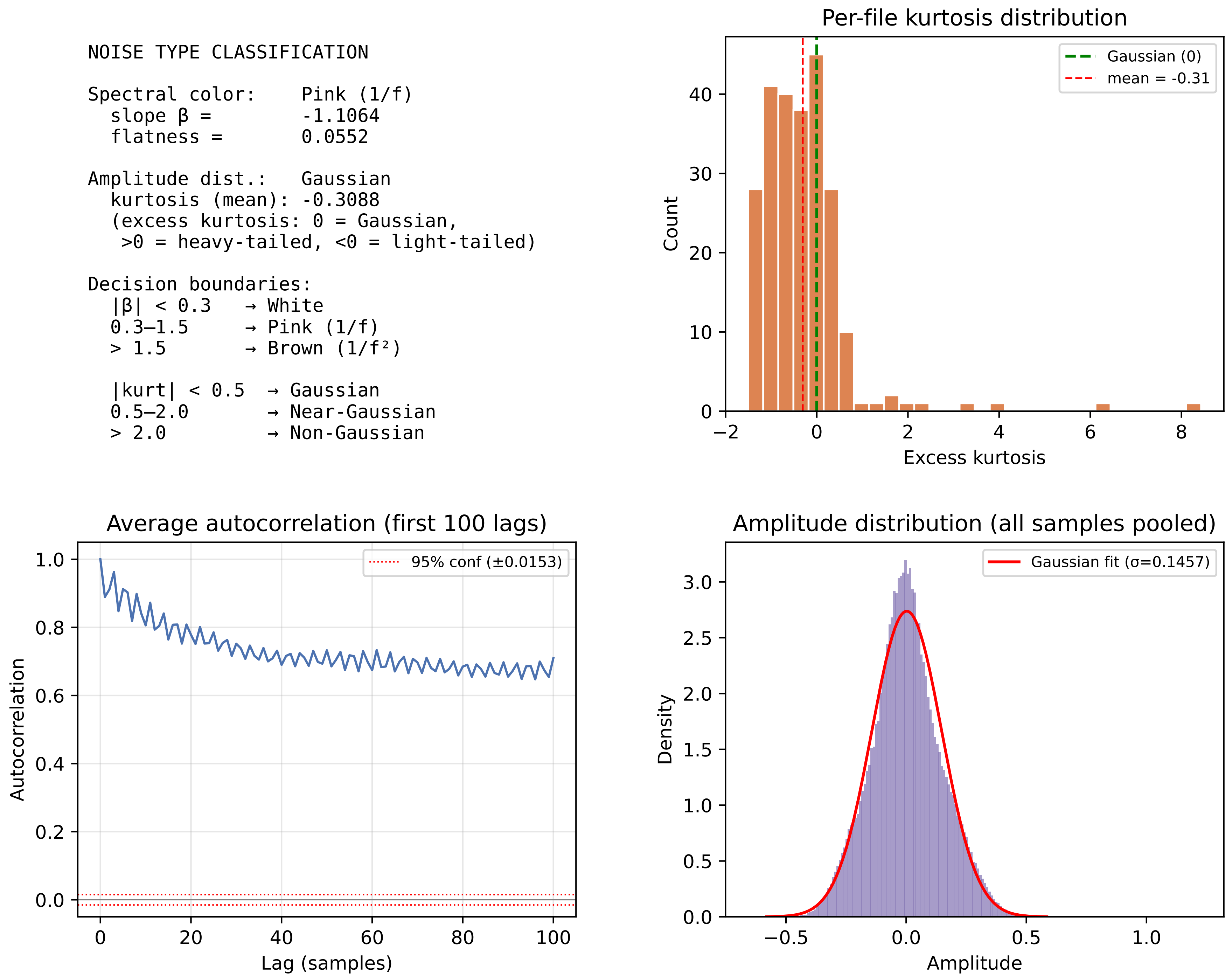




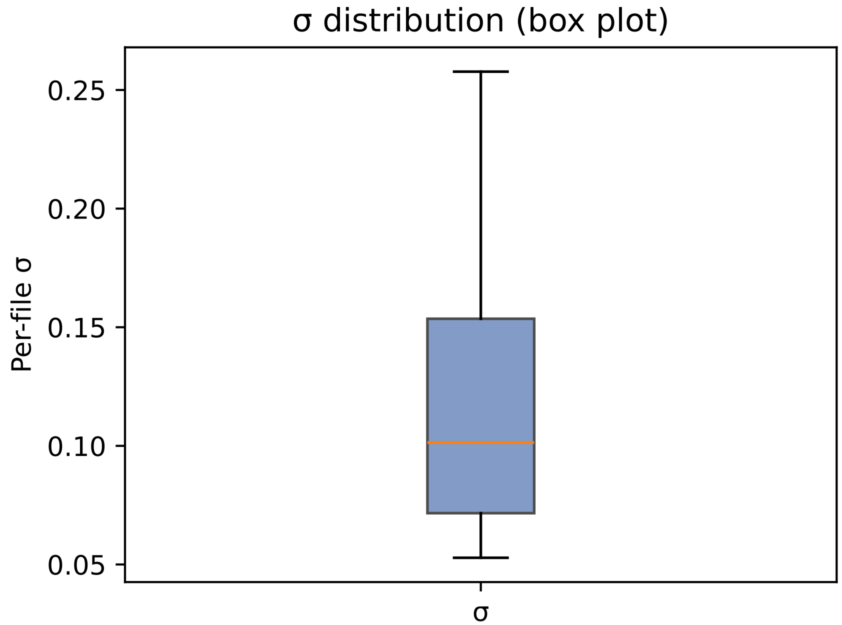
BAND ENERGY SUMMARY

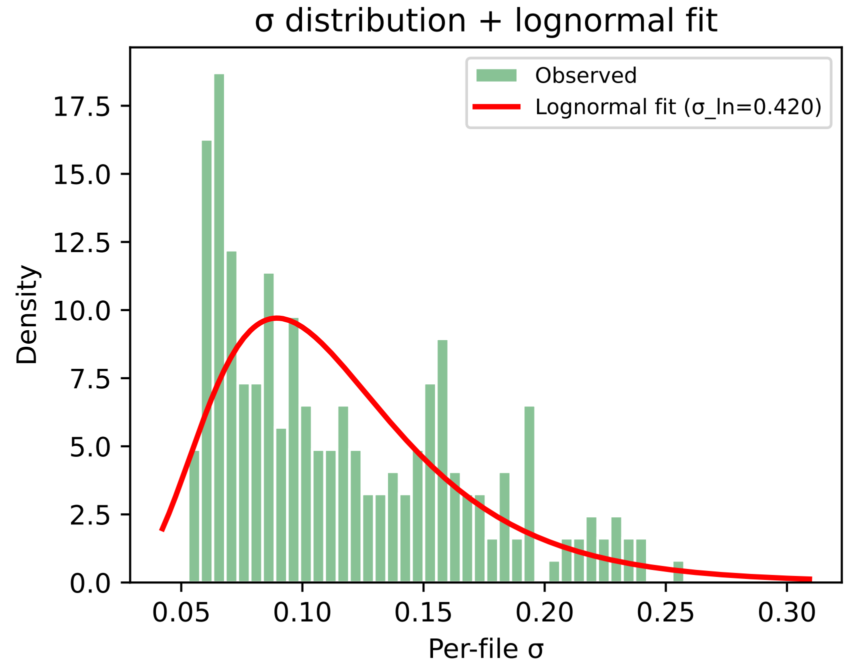
Band	Mean %	Std %
0–1 kHz	1.7%	0.6%
1–7 kHz	20.9%	4.8%
7–10 kHz	9.8%	7.2%
10–40 kHz	34.2%	11.5%
40–80 kHz	8.8%	1.9%
80 kHz–Nyq	26.2%	5.5%
In-band [7–80k]	49.5%	9.2%

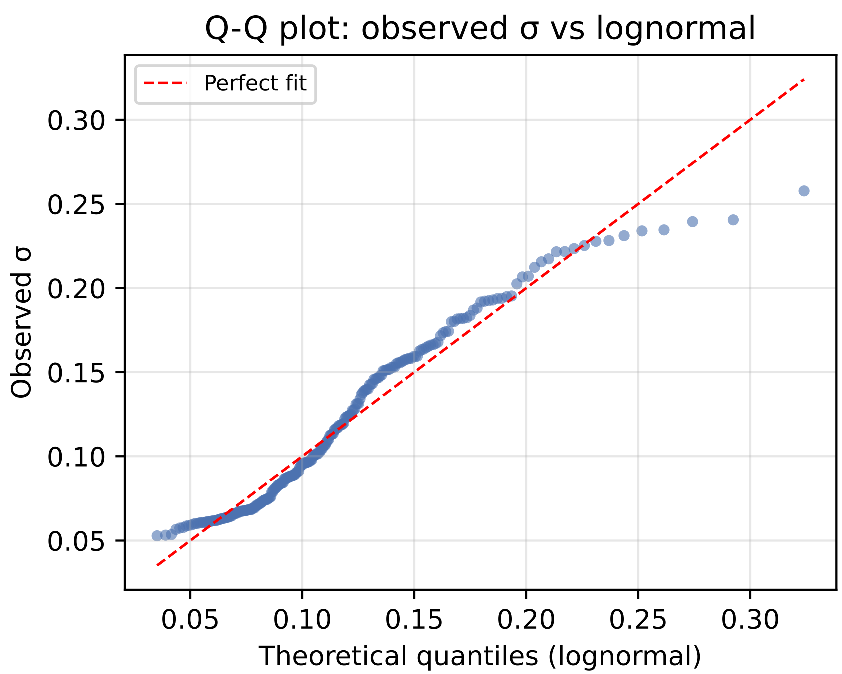
Noise Type Classification — 2um



Amplitude Variability (Inter-file) — 2um





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AMPLITUDE VARIABILITY SUMMARY

Number of files:     240
mean(σ):             0.116761
std(σ):              0.050236
CV = std(σ)/mean(σ): 0.4302

Lognormal fit parameters:
  shape (σ_ln):      0.4205
  scale (exp(μ_ln)): 0.106793

Reference (from real noise files):
  CV_ref = 0.19
  Ratio CV/CV_ref:   2.26

Interpretation:
  High variability — mixed conditions or signal content
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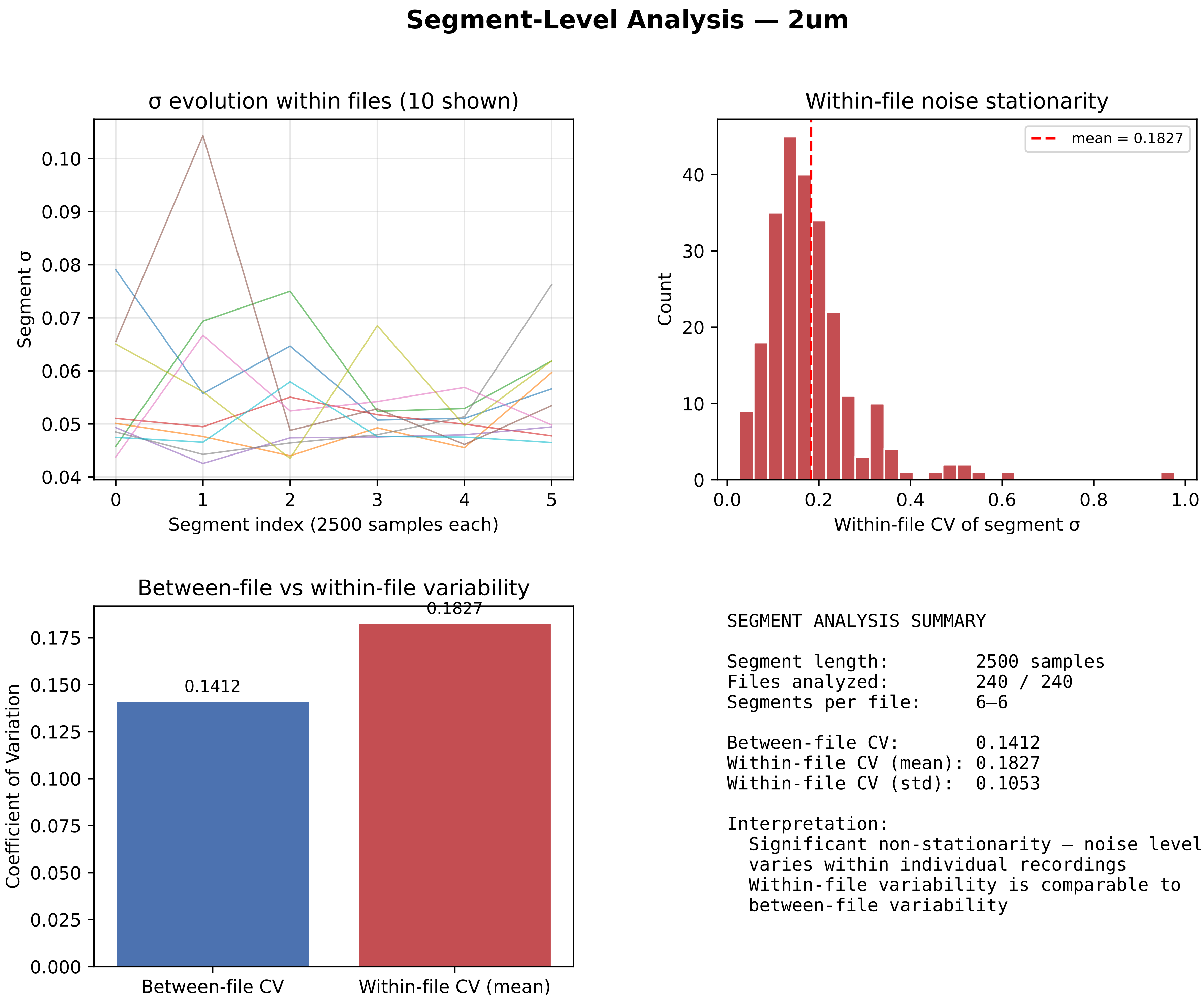

Segment-Level Analysis — 2um
σ evolution within files (10 shown)
Segment σ
Segment index (2500 samples each)
0.04
0.05
0.06
0.07
0.08
0.09
0.10
0
1
2
3
4
5
Within-file noise stationarity
mean = 0.1827
Count
Within-file CV of segment σ
0
10
20
30
40
0.0
0.2
0.4
0.6
0.8
1.0
Between-file vs within-file variability
Coefficient of Variation
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.1412
0.1827
Between-file CV
Within-file CV (mean)
SEGMENT ANALYSIS SUMMARY
Segment length: 2500 samples
Files analyzed: 240 / 240
Segments per file: 6–6
Between-file CV: 0.1412
Within-file CV (mean): 0.1827
Within-file CV (std): 0.1053
Interpretation:
Significant non-stationarity — noise level
varies within individual recordings
Within-file variability is comparable to
between-file variability